

Autonomous Driving Sensor Data Report

Detailed Breakdown of matlab_data_2 Export

1. Overview & Data Source

This report details the structural analysis of the 'matlab_data_2' directory. The data originates from a MATLAB Automated Driving Toolbox simulation, consisting of three main sensors: Radar, Camera, and LiDAR. Each sensor's output has been meticulously analyzed and reformatted into Python-accessible structures.

2. Radar Data (r_data.mat)

Format: MATLAB Struct (.mat)

Contents: 50 independent radar detections.

Fields Extracted:

- Time: Simulation timestamp of detection.
- SensorIndex: Unique ID of the radar sensor.
- ObjectClassID: Numerical classification of the detected object.
- Measurement: Numerical array typically containing [Azimuth, Range, Range_Rate].

Interpretation: The 'Measurement' field provides crucial spatial and velocity metrics (Range and Range_Rate) which are used to calculate the Time-to-Collision (TTC) for risk scoring algorithms.

3. Camera Data (c_data.mat)

Format: MATLAB Struct (.mat)

Contents: 50 independent camera object detections.

Crucial Note: Unlike traditional camera exports, this file does NOT contain raw RGB pixel arrays or video frames. The simulation exported processed 'Camera Detections' (e.g., bounding boxes and class IDs). Because it lacks raw pixels, image frames cannot be reconstructed.

Fields Extracted:

- Identical structure to radar data, containing Time, SensorIndex, ObjectClassID, and Measurement.
- Measurement typically contains 2D Bounding Box coordinates [x, y, width, height].

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4. LiDAR Data (Lidar_PCD_Data/)

Format: Standard Point Cloud Data (.pcd) Files

Contents: 800 individual frames of 3D spatial points.

Data Dictionary (Per Point):

- X, Y, Z: Euclidean spatial coordinates in meters relative to the ego-vehicle.

Interpretation: Because these are standard .pcd files, they can be natively ingested by Python using the 'open3d' library. To utilize this data for trajectory prediction, the points must first undergo RANSAC Ground Plane Removal, followed by DBSCAN clustering to group points into distinct 3D objects.

5. Conclusion & Actionable Steps

The proprietary MATLAB files have been successfully unpacked. The radar and camera detections have been extracted into clean CSV spreadsheets located in the 'Results/' directory. The 800 LiDAR .pcd files are ready for immediate Python processing via Open3D.